

**SMART WOUND MONITORING: AI-DRIVEN IMAGE
ANALYSIS BASED ON THE T.I.M.E. FRAMEWORK**

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ON THE T.I.M.E. FRAMEWORK**

Field of Study: **DEEP LEARNING, AI IN HEALTHCARE**

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TABLE OF CONTENTS

Table of Contents	iii
List of Figures	vi
CHAPTER 1: INTRODUCTION.....	1
1.1 Background.....	1
1.2 Problem Statement.....	3
1.3 Research Objectives.....	4
1.4 Research Questions.....	4
1.5 Scope of Study	5
1.6 Significance of Study.....	5
1.7 Organization of the Report	6
CHAPTER 2: LITERATURE REVIEW.....	7
2.1 Introduction.....	7
2.2 Wound Management and Assessment	8
2.2.1 Wound Classification	8
2.2.2 Factors Affecting Wound Healing	8
2.2.3 Wound Bed Principles	9
2.2.4 TIME Framework for Wound Assessment.....	9
2.2.5 Challenges with TIME Framework	10
2.3 Digital and Image-Based Wound Assessment.....	10
2.4 Tissue Segmentation and IME Classification Pipeline.....	12
2.4.1 Tissue Segmentation.....	12
2.4.2 IME Classification.....	12
2.4.3 Integration of Segmentation and Classification	13

2.4.4	Address Label Quality and Variability	14
2.4.5	Evidence from Wound-Specific Studies	14
2.5	Deep Learning Models for Medical Image Classification.....	14
2.6	Mobile Deployment of Medical Analysis Models	16
2.7	Research Gap	17
CHAPTER 3: METHODOLOGY		19
3.1	Overview of Proposed Framework	19
3.2	Dataset Description.....	20
3.3	Data Preprocessing	21
3.3.1	Image Standardization	21
3.3.2	Region of Interest (ROI) Extraction	21
3.4	Exploratory Data Analysis (EDA).....	22
3.4.1	IME Class Distribution.....	22
3.4.2	Image Quality and Variability	23
3.5	Tissue Segmentation (T-SegNet).....	23
3.5.1	Role of T-SegNet.....	23
3.5.2	Integration and Scope	23
3.6	IME-ClassNet Model Architecture	24
3.6.1	Multi-Task Learning Design	24
3.6.2	Convolutional Backbone Network	24
3.6.3	Task-Specific Classification Heads.....	25
3.7	Model Training	25
3.7.1	Supervised Transfer Learning and Fine-Tuning.....	25
3.7.2	Data Augmentation.....	26
3.7.3	Handling Class Imbalance	26
3.7.4	Optimization Configuration	26

3.8	Semi-Supervised Learning.....	26
3.8.1	Baseline Model Training	26
3.8.2	Pseudo-Label Generation and Filtering.....	27
3.8.3	Model Retraining.....	27
3.9	Model Evaluation.....	27
3.9.1	Stratified k-fold Cross-validation.....	27
3.9.2	Evaluation Metrics	28
3.9.3	Reporting	29
3.10	Model Deployment and System Evaluation	29
3.10.1	Model Deployment Framework	29
3.10.2	End-to-End Performance Benchmarking	29
3.10.3	Performance Target and Evaluation Criteria.....	30
3.11	Summary.....	30
	References	31

LIST OF FIGURES

Figure 1.0: Human skin layers	1
Figure 2.0: Overview of the proposed TIME-Net Framework	20
Figure 3.0: Class Distribution for Infection, Moisture, and Edge tasks	22

CHAPTER 1: INTRODUCTION

1.1 Background

The outermost layer of human body is the skin layer. It consists of multiple layers; the epidermis, dermis, and subcutaneous tissue as shown in Figure 1.0. Its function is to protect our internal organs and also serves as our first line of defense, protect us against UV rays, diseases, and mechanical harm (Yousef et al., 2024).

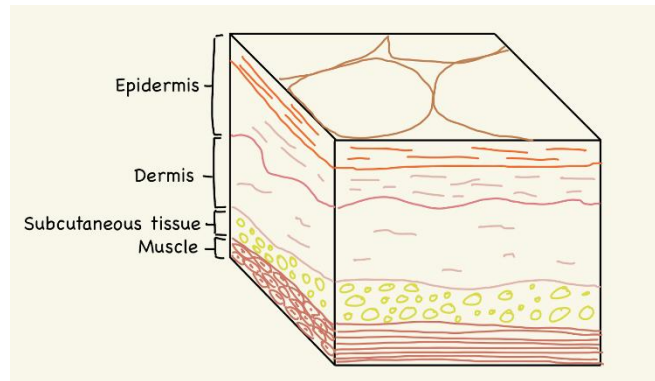


Figure 1.0: Human skin layers, which consist of epidermis, dermis, and subcutaneous tissue (Wehman, 2022)

A wound is formed when the integrity of the skin barrier is compromised by an injury (Afonso et al., 2021). Wounds can be categorized into acute and chronic types based on healing rate and process (Tsegay et al., 2022). Acute wounds usually recover following the normal healing phases, starts from hemostasis, inflammation, proliferation, to remodeling within a predictable timeframe. Whereas, chronic wounds always cannot heal within the expected timeframe (Wallace et al., 2023). Diabetic foot ulcers, venous, arterial, and pressure ulcers are the four types of chronic wounds (Falanga et al., 2022). The treatment plan may differ for each wound type (Boulton, 2020; Robles-Tenorio et al., 2022; Grey et al., 2006; Zaidi & Sharma, 2024).

A study conducted in German reported that the prevalence of chronic wounds was about 1-2% in the general population (Heyer et al., 2016). Some chronic wounds persist for many years and sometimes for decades, which increases the risks of secondary complications such as depression and social isolation. According to the statistical study,

the mortality rate after the development of a diabetic foot ulcer was significantly high at 40% (Jupiter et al., 2015). Elderly patients are in higher risk because their skin is thinner and less elastic. Without a proper wound care and management, chronic wounds will easily develop into infection and hospitalization (Shim et al., 2025).

A proper wound assessment is essential to help the wound recover effectively. Clinicians or healthcare professionals need to review some important characteristics of the wound at each review session. These include wound size, tissue type, presence of infection or inflammation, moisture level, and the condition of the wound edge (Mustafa Alhababi et al., 2024). The TIME framework is one of the frequently used assessment tools in wound care management. The 4 key components of TIME are Tissue, Infection or Inflammation, Moisture, and Edge. Tissue component focuses on assessing the wound composition, the non-viable tissue will be removed, while the healthy granulation tissue will be preserved. Infection or Inflammation component addresses bacterial burden. The Moisture component relates to the control of exudate so that the wound environment is not too wet or too dry. The last component, Edge, refers to the epithelial migration from the wound edge. This provides clinicians with a consistent language to describe wound conditions. This allows the development and coordination of effective treatment plans across multiple care settings (Amani et al., 2023).

Digital technology is used more frequently in daily life. According to recent studies, almost two-thirds of people worldwide, including elderly, own a smartphone. From a wound care perspective, clinicians and patients often use smartphones to capture wound photographs as a record. It also enables remote review when frequent clinic visits are difficult. With this, it creates an opportunity to develop a smart wound assessment tool (Blackburn et al., 2019).

1.2 Problem Statement

Wound management is a very important practice in the wound healing process. Patients with chronic diseases need frequent check-ups in order to track their wound healing progress, clean the wound, and get appropriate treatment before it leads to other complications. However, many of these patients have limited mobility and is not convenient for them to travel for a check-up (Amani et al., 2023).

The TIME framework is commonly used as a practical tool for wound care assessment, which focuses on 4 components: Tissue, Infection/Inflammation, Moisture, and Edge of the wound. The traditional wound assessment is purely done manually by healthcare professionals. They usually observe the wound with their naked eyes and record all the observations manually. This can be very time-consuming as each wounds need to be evaluated one by one. This practice also relies heavily on the clinician's experience and different clinicians may have different descriptions and judgements, which can lead to inconsistent decisions about wound dressings and treatments. These inconsistencies can also affect the healing process of the patient (None Rishi Kesaraju, 2025).

With the advancement of hardware and imaging tools, wound photographs are now commonly used in clinical practice and research (Chen et al., 2024). Image processing and deep learning techniques also used broadly to automate wound assessment process partially, such as wound segmentation or tissue types annotation. However, most of them are still focusing on isolated tasks, rather than providing full TIME-based wound annotation (Moortgat et al., 2024). This study aims to develop and evaluate a deep learning wound assessment framework that combines a tissue segmentation model and a multi-task IME classifier. Then, examine the system's accuracy and practicality for near real-time use in a smartphone-based client application.

1.3 Research Objectives

This research aims to design and evaluate a smart wound assessment framework, known as TIME-Net. This framework consists of 2 deep learning models, which are a tissue segmentation model (T-SegNet) and a multi-task IME classification model (IME-ClassNet).

The objectives of this research are:

1. To develop the IME-ClassNet model, a multi-task deep learning framework for the classification of Infection, Moisture, and Edge of chronic wounds.
2. To evaluate IME-ClassNet performance on a wound image using F1-score and Sensitivity metrics to ensure reliable detection of pathological features.
3. To integrate the IME-ClassNet model, with a tissue segmentation model, T-SegNet, into a smartphone-based prototype, and to evaluate the overall response time.

1.4 Research Questions

To fulfil the research objectives stated above, this study focuses on the following research questions:

1. How effectively does the multi-tasks IME-ClassNet model perform Infection, Moisture, and Edge classification in chronic wound images?
2. What is the diagnostic performance of the proposed IME-ClassNet model in term of F1-score and Sensitivity metrics?
3. Can the trained TIME-Net, integrating both T-SegNet and IME-ClassNet models, achieve feasible end-to-end response times when deployed as a cloud-based smartphone workflow?

1.5 Scope of Study

The scope of this study is to develop an automated smart wound assessment tool under the TIME framework, known as TIME-Net. The dataset is obtained from both public sources and private clinical collections. All wound images will be anonymized and used only for research purposes.

The core analysis is performed by two sub-models: a tissue segmentation model (T-SegNet) and a classification model for Infection, Moisture, and Edge (IME-ClassNet). A pre-trained segmentation model provided by a peer researcher will be used to obtain the wound masks as the input of this study. The study methodology covers data preparation, model training, testing, and performance evaluation using standard metrics, including Intersection-over-Union for tissue segmentation and accuracy, precision, recall, and F1-score for classification.

The study also includes the integration of trained models into a mobile application prototype to achieve near-real-time use within a smartphone-based workflow. This focuses on the inference performance and basic display of TIME-related outputs. This study does not include a full clinical trial, long-term outcome tracking, or treatment recommendations. The tool is designed to support, rather than replace, clinical judgement.

1.6 Significance of Study

This study supports more structured and consistent wound assessment in clinical practice. By providing the TIME-based outputs from wound photos, it can provide clinicians with extra objective information instead of relying only on manual visual judgement. Besides, this study also benefits patients with limited mobility. They can assess their wound with just a single capture of their wound using a smartphone at home or in community settings. Overall, this study offers a framework that links wound masks, tissue segmentation, IME classification, and smartphone-based deployment via cloud-

hosted inference services. This framework serves as a reference for future image-based decision support tools in wound care.

1.7 Organization of the Report

This report is organized into 3 chapters. Chapter 1 is the introduction of the study. It consists of the study background, problem statements, research objective, research questions, scope, and significance of the study. Chapter 2 provides the literature review. It discusses factors that affect wound healing for different types of wounds. Then compare existing methods for image-based wound assessment and related work on deep learning, tissue segmentation, IME classification, and mobile-based wound applications. This chapter will also highlight the research gap that this study aims to address. Chapter 3 emphasizes the research methodology. It explains the overall system design of TIME-Net and the main steps carried out to achieve the objective of this research.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Wound care and assessment are important practices in wound management and the healing process, especially for patients with chronic wounds, which take longer than expected time to heal and do not heal quickly on their own (Nagle et al., 2023). Frequent wound check-ups are necessary to prevent the wound getting worst and lead to other complications. Traditional wound assessment is done manually and heavily relies on clinician's expertise. Different clinicians may describe the wound differently (Griffa et al., 2024). This can be a major issue as inconsistent description could affect the decision for future treatment plans and subsequently delay the healing progress. The TIME framework provides a structured and standard for wound assessment. It categorizes wound assessment into 4 key components, which are Tissue, Infection/ Inflammation, Moisture, and Edge of the wound. This acts as a guideline for clinicians to assess these four key features of the wound at each review and supports more systematic decision-making. However, TIME-based assessment is still performed manually in most clinical settings, thus, variability still occurs. This limitation should not be ignored because consistent evaluation supports better treatment choices and clearer follow-up on wound healing progress.

Digital imaging has expanded image-based wound assessment in both clinic and home settings. Patient can easily assess their wound with just a single capture of wound images using a smartphone device. With this, they can monitor their wound condition and help in easy documentation anywhere and at any time. This automated wound analysis supports telehealth workflows when in-person review is difficult. Deep learning has shown strong performance in medical image segmentation and classification tasks (LUCAS et al., 2020). This chapter reviews published work on wound assessment, digital and image-based approaches, tissue segmentation and TIME classification, deep learning

models, integrated frameworks, and mobile deployment, then synthesizes the research gap that motivates TIME-Net.

2.2 Wound Management and Assessment

2.2.1 Wound Classification

There are two types of wounds, which are Acute and Chronic. Wounds are classified based on the severity and healing progress. Wound healing is a very complex biological process that go through 4 overlapping phases: hemostasis, inflammation, proliferation, and remodeling (Wilkinson & Hardman, 2020). No matter acute or chronic wound, both go through the same sequence, but chronic wounds normally takes longer time in inflammation phase, or impaired repair (Schultz et al., 2011).

2.2.2 Factors Affecting Wound Healing

Some key drivers are causing the wound to stall, especially in oxygen-deficient environment. Local hypoxia reduces tissue repair, causing the wound to be stalled in the inflammation phase. Diabetes and vascular disease reduce perfusion and impair immune response and inhibit the tissue regeneration process. Venous hypertension increases oedema and fluid burden. Prolonged pressure reduces blood supply and causes tissue breakdown (Guo & DiPietro, 2010).

A clinical review of wound healing highlighted patient lifestyle and wound characteristics that affect healing results, such as oxygenation, infection burden, diabetes, nutrition supply, smoking behaviors, medication use, etc. (Guo & DiPietro, 2010). According to Aakansha Giri Goswami et al. (2023) and Almuhanha (2025), microbial biofilms cause chronic wounds to heal more slowly by causing persistent inflammation and reduced response to treatment.

2.2.3 Wound Bed Principles

Wound bed preparation principles will be applied based on scenarios, such as removal of necrotic tissues, control the condition of infection, balance exudate level, and support the edge advancement (Schultz et al., 2004). The wound bed preparation is not just to cover the wound, but to prepare the wound bed by correcting all the underlying imbalances (Sibbald et al., 2021).

2.2.4 TIME Framework for Wound Assessment

In early 2000's, Schultz et al, (2003) introduced TIME framework as a practical tool to standardize clinician reviews and to link findings to practical actions. It focused on 4 specific barriers to healing, which are Tissue, Infection/ Inflammation, Moisture, and Edge of wound. Each component requires specific visual observations that determine the next treatment plan (Schultz et al., 2003). The first component, Tissue, focuses on the wound bed composition, which can be classified into necrotic tissue, slough, and granulation. It acts as an indicator to help clinicians differentiate between viable (healthy red granulation tissue) and non-viable tissue (yellow slough and black necrotic tissue). Debridement is required when the non-viable tissues are identified, as it will block the healing process (Garten et al., 2020). For Infection of Inflammation, it focuses on bacterial burden or host response. The classic signs and symptoms can be summarized as NERDS (Non healing, exudate increase, red friable tissue, debris, and smell), surrounding redness, purulence, or cellulitis. Moisture balance covers the wound fluid environment (Orsted et al., 2018). Excess exudate and periwound maceration indicate too wet condition. Whereas dryness and adherent tissue suggest insufficient moisture. The class between these two is considered moderate moisture, which is the best condition for the wound to be repaired. The fourth criterion is the edge of the wound, which focuses on epithelial advancement and margin condition. Conditions like rolled edges, callus, and

poor periwound, are an indication for stop healing (The Royal Children's Hospital Melbourne, 2023).

2.2.5 Challenges with TIME Framework

Other than the standardization of TIME framework, its application in routine clinical practice remains heavily dependent on subjective visual inspection. For example, measurement of wound size is often done by using a ruler, tracing, or digital planimetry (A Comparison of Computer-Assisted and Manual Wound Size Measurement, 2026). Besides, the classification of tissue type, sign of infection, exudate level, or edge advancement is all observed using the naked eye. Studies such as Vestjens et al. (2017), have shown that different clinicians frequently disagree on tissue classification. They got low agreement between nurses when determining the percentage of slough over granulation tissue of a wound. This further support the need of a more objective and repeatable assessment tool, especially when the consistent reporting is required under structured TIME framework.

2.3 Digital and Image-Based Wound Assessment

Wound photographs is the main component for digital wound assessment in order to support documentation, follow-up, and remote reviews. McLean et al. (2023) study showed that remote digital wound monitoring tool has been implemented in 2 UK Hospitals. They prove that remote wound assessment is applicable within a real care pathway. Similar findings from Gunter et al. (2016) also demonstrate that high usability and acceptance of smartphone-based wound monitoring when patients are guided for image capture and reporting. These approaches successfully reduce the need of frequent in-person visits and help track wound status across time.

Image quality is the key requirement for any image-based assessment. If the image is poor in quality, such as insufficient lighting, incorrect angle, distance, blurriness, and the

absence of a scaling reference object, it may affect the annotation results from both human and automated analysis. Zhang et al. (2021) developed a home-based mobile health tool that included real-time feedback to help patients capture usable wound images, and they reported that this feedback improved overall image quality while supporting home monitoring. This prove that structured capture guidance is important and needed for a digital workflow to provide accurate and reliable assessment.

Digital wound assessment also includes measurement and tracking tools. Digital methods aim to improve consistency in wound size and feature recording compared with manual measurement. A systematic review evaluated digital devices for wound measurement and highlighted ongoing variation in accuracy and reporting across tools, which shows the need for careful validation and standard measurement practice (Kientzy et al., 2024). In the same direction, published work continues to propose practical solutions to reduce measurement error from photo-taking variation. They have added reference markers to support consistent scaling (Li et al., 2025). These studies support the value of digital methods, while also showing that capture conditions and standardisation still shape reliability.

Automated features are currently available in a large number of commercial and academic applications; however, the strength and quality of the data differ between applications. Kabir et al. (2024) evaluated wound assessment apps and discovered gaps in the evidence-based reliability of existing apps. Another evaluation of AI-integrated mobile apps for ulcer segmentation reveals that many systems still focus on a particular task, primarily segmentation, rather than a complete clinical framework (Griffa et al., 2024). This creates a need for existing research that links multi-component evaluation, segmentation, and image capture under a structured model like TIME with performance and usability evaluated for mobile devices.

2.4 Tissue Segmentation and IME Classification Pipeline

2.4.1 Tissue Segmentation

Tissue segmentation is always the start for image-based wound assessment. This is because the model needs to first separate the wound from the surrounding skin and background. This step is important as a wound-focused input can reduce noise from any irrelevant background objects (Veredas et al., 2015). Previous studies used traditional image processing methods such as thresholding, clustering, and contour-based techniques to show the wound boundaries (Anisuzzaman et al., 2021). However, these methods often perform poorly when image conditions vary, such as lighting, skin tone, camera angle, or wound appearance (Ahmad Fauzi et al., 2015).

Deep learning segmentation reduces reliance on hand-made rules by learning wound features directly from labelled images. Many studies use encoder-decoder designs, often based on U-Net, to produce pixel-level wound masks. After that, they also evaluate the result using IoU or Dice (Cassidy et al., 2021). A challenge report on diabetic foot ulcer segmentation also describes the evaluation of localization, classification, and segmentation methods under shared rules, which shows that wound segmentation has matured into a benchmarked task (Yap et al., 2024).

2.4.2 IME Classification

The IME classification covers infection or inflammation, moisture balance, and edge of wound. Infection or Inflammation classification often receives more attention because infection links to risk and escalation. A review of machine learning approaches for image-based identification of surgical wound infections summarized published image-based methods and stressed the need for stronger validation across settings (Pablo et al., 2024). For moisture and edge, literature often treats them indirectly through tissue and periwound cues, or through segmentation outputs, because these features can be subtle

and image quality can change them. Chang et al. proposed a super pixels approach to help tissue labelling for classification, which enhances tissue feature extraction from wound images. Besides, Chairat et al. (2023) demonstrated that smartphone images can allow multi-region outputs when labels are provided by using deep learning methods to segment wound area and tissue regions.

2.4.3 Integration of Segmentation and Classification

In medical image analysis, segmentation and classification are often independent tasks. Segmentation isolates the wound region, also known as the region of interest, while classification assigns a diagnostic label to that region. There are two architectural strategies that can be used to integrate both segmentation and classification simultaneously, which are a sequential two-stage pipeline or a unified multi-task network.

In sequential two-stage pipeline, segmentation will first isolate the region of interest, then classification will predict labels from the segmented region. A breast ultrasound study described a “segmentation-guided classification” pipeline, where lesion segmentation acts as the input for the classification stage for diagnosis, showing the two steps work together as one decision workflow (Inan et al., 2022). Similar two-stage designs also appear in broader medical imaging, where a first stage localizes or segments the target and a second stage performs diagnosis. This is very helpful, especially when background dominates the image, as this can cause bias classification (Abdel-Basset et al., 2021).

Another design is the multi-task network, which merges segmentation and classification in a single model. In this approach, one network shares early feature layers and produces both a pixel-level mask and a classification output. A dual-branch skin lesion study reported a framework that performs segmentation and classification together, which supports unified training and consistent feature learning for both outputs (Owida

et al., 2025). This model does not require two models to run, but it requires careful training because both outputs must learn well at the same time.

2.4.4 Address Label Quality and Variability

Label quality is a key limitation for both segmentation and IME classification. Image labels often come from visual judgement, and agreement is not equal across tissue types. A deep learning study on wound tissue segmentation reported inter- and intra-rater variability during annotation and treated this as a practical barrier for objective assessment. This is important for IME tasks because infection, moisture, and edge labeling all rely on visual indicators that are influenced by experience and lighting (Ramachandram et al., 2022). These results support the study's approach, which focuses IME characteristics on the wound region using a segmentation output and assesses performance using standard metrics for both segmentation and classification.

2.4.5 Evidence from Wound-Specific Studies

Multi-task pipelines reduce duplication and can share useful features, but they need good labels for both tasks and careful balancing of losses during training. Wound-specific evidence also supports combined workflows. A pilot study trained deep learning models for wound segmentation and wound classification on the same image set, showing that both tasks can be handled within one study design, while also reflecting the practical need for reliable segmentation and labels when classification depends on wound region cues (Zimmermann et al., 2023).

2.5 Deep Learning Models for Medical Image Classification

The limitation of traditional image processing has brought the adoption of Deep Learning. Convolutional neural networks (CNNs) are a primary model family use for many medical image tasks, such as segmentation, detection, and classification (Litjens et al., 2017; Cai et al., 2020). They also summarize common architectures such as AlexNet,

VGG, GoogleNet, and ResNet. These models often appear in 2 practical ways. First way is to train a network from scratch when a large, labelled dataset exists. The second way is transfer learning, which starts from a pre-trained model on a large image dataset, then transfer the knowledge to the smaller dataset (Esteva et al., 2019). This technique is more frequently applied in medical dataset, as medical data usually are small and limited (Shen et al., 2017).

Many medical problems also involve more than one label or more than one related task. Multi-task deep learning addresses this by learning shared features for related outputs, then producing separate predictions for each task. This technique align with our IME model, this model also using a shared feature extractor and covers 3 wound features. In practice, multi-task models often share early layers for general visual features, then branch into task-specific heads near the end of the network. The goal is improved learning efficiency and better generalisation when tasks share visual cues (Shim et al., 2025).

Model evaluation in medical image classification relies on the metrics that match the specific clinical goal and the class balance of the dataset. Accuracy alone can hide poor performance on minority classes when classes are imbalanced. Therefore, matrices like precision, recall, and F1-score are often reported because they describe different types of errors and reflect the false positives and false negatives (Litjens et al., 2017). Macro-averaged scores also help reflect performance across all classes rather than just bias towards the domain class. Confusion matrices also help explain which categories are most often confused, which supports error analysis and label refinement (Garcia et al., 2015).

Small dataset and class imbalance are the two commonly seen challenges in medical image classification studies. A small dataset will increase overfitting risk, especially when image capture conditions are different across hospitals or devices. Class imbalance occurs when severe cases occur less often than normal cases. This can lead to biased

towards the majority classes when training images are different from deployment images in terms of lighting, camera type, patient group, or clinical settings (Cheplygina et al., 2019). Transfer learning, data augmentation, class-weighted losses, resampling techniques, and strong validation designs are some popular approaches (Morid et al., 2021). Semi-supervised learning also provides a solution by increasing the volume of data by using both labelled and unlabeled datasets (Liu et al., 2023). Reviews also emphasize the importance of clearly describe and evaluate data from many sources in order to show generalization (Zech et al., 2018).

2.6 Mobile Deployment of Medical Analysis Models

Mobile deployment refers to running medical image analysis within a smartphone-based workflow, where the smartphone is used to capture the image. There are two ways to execute model inference, which are on-device (edge inference) and remotely (cloud-hosted inference). Edge inference supports faster feedback because it reduces reliance on uploading images to a remote server. However, studies on mobile deep learning show that deployment on smartphones faces practical limits in terms of processing power, memory, battery use, and heat, so model design must strike a balance between accuracy, speed, and resource consumption (Wang et al., 2021). Therefore, studies often choose to use lighter model families and apply some optimization including the use of compact backbones, reduce input resolution, and apply compression to shrink the model size in order to speed up the inference time. A wound assessment tool should return results fast for routine treatment and home monitoring, but the input resolution cannot be reduced, as it will affect the result of wound annotation.

On the other hand, because cloud-hosted inference removes the need for runtime integration and mobile-specific model conversion, it is commonly used in prototype-stage systems. This approach relies on a smartphone application to send the input wound

images (or extracted regions of interest) to a managed inference endpoint, which then uses an API to return segmentation and classification results. This design supports rapid iteration and centralized updates of model versions.

In this study, the TIME-based pipeline is integrated into a smartphone-based prototype that performs inference through a managed cloud inference service accessed via API. The integration follows a two-stage workflow where tissue segmentation output guides ROI extraction, followed by multi-task IME classification. The performance is evaluated by measuring the end-to-end response time of the cloud-based pipeline to assess its feasibility for practical wound monitoring use.

2.7 Research Gap

Previous studies confirm that deep learning can support wound image analysis, especially for wound segmentation and tissue-type classification. This progress supports more objective wound measurement and improves consistency compared to manual justification. The TIME framework is widely used in clinical wound care, but it is still performed manually and depends on visual judgement. This can lead to subjectivity and variability, which will also affect consistency in treatment decisions.

There are three gaps that remain to be filled in this study. Firstly, current work lacks holistic TIME-based assessment. Most deep learning studies focus on isolated tasks, mainly wound segmentation or tissue classification, which addresses the “Tissue” component of TIME. There are lacking of studies focus on Infection or Inflammation, Moisture, and Edge of the wound within a TIME-based assessment workflow. Current tools do not provide complete multi-task model that reflect routine TIME-based clinical review (Morgado et al., 2025).

Secondly, there are lacking multi-task models that focusing on all 3 Infection/Inflammation, Moisture, and Edge (IME) features. Segmentation and clinical feature prediction are always treated as separate steps or implemented as separate models. This increases computational cost and system complexity. There are still a need for an integrated pipeline where segmentation focuses the region of interest and a shared-feature multi-task classifier predicts the IME outputs together, using consistent inputs (Mohammed et al., 2025).

Thirdly, deployment feasibility remains unclear for complete TIME-style pipelines. Smartphone-based wound photography is common, but many deep learning models are developed and evaluated in desktop or server environments. Few studies report end-to-end inference performance for combined segmentation and multi-attribute IME classification on standard smartphone devices. These limits understanding of whether such systems can achieve near-real-time inference particularly when inference is provided through managed cloud services accessed via APIs (Anisuzzaman et al., 2021).

Therefore, this study addresses these gaps by proposing TIME-Net, an integrated deep learning framework that combines a tissue segmentation model (T-SegNet) with a multi-task IME classification model (IME-ClassNet). Then, evaluate the analytical performance and deployment feasibility to support structured, objective, and accessible TIME-based wound assessment.

CHAPTER 3: METHODOLOGY

3.1 Overview of Proposed Framework

This study proposes an automated wound assessment framework known as TIME-Net. It is designed to support wound assessment based on the TIME framework using wound images captured by smartphone devices. The main purpose of developing TIME-Net is to help provide a more structured and consistent evaluation of chronic wounds by analyzing key features related to Tissue, Infection or Inflammation, Moisture, and Edge conditions.

TIME-Net is the workflow that consists of 2 main components: a tissue segmentation model, T-SegNet, and a multi-task classification model, known as IME-ClassNet. The T-SegNet functions to segment out the wound area and provide a wound mask that helps the analysis focus on the wound itself and reduce the background noise. The segmented wound images will then be passed to the IME-ClassNet model to predict the Infection or Inflammation, Moisture, and Edge condition at the same time.

In this study, the T-SegNet model is developed by a collaborator and is used as a pre-trained model. It will only be used to generate the wound masks as the input for IME-ClassNet model. The design, training, and optimization of T-SegNet are not part of the scope of this study. Instead, this research fully focuses on how the segmented output can be used to support more accurate and consistent IME classification.

The main contribution of this study still lies in the development and evaluation of IME-ClassNet, and also the integration of both models into a single TIME-based assessment framework that can be used on a smartphone device. The main goal is to achieve reliable classification performance with limited labelled data and ensure the overall system can produce results within a short time that are suitable for practical use. The overall TIME-

Net workflow, from wound image input to TIME-based result output is shown in **Figure 2.0**.

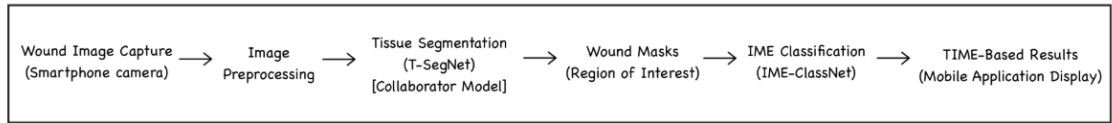


Figure 2.0: Overview of the proposed TIME-Net Framework

3.2 Dataset Description

This study uses a hybrid dataset that combines publicly available wound images and private clinical wound images. This approach is chosen to increase the variety of wound appearances and to better reflect real-world wound assessment conditions. All the images are only used for research purposes. Any private clinical images are anonymized to ensure that no patient-identifiable information is included.

The publicly available dataset is obtained from Kaggle, which was compiled from multiple published wound segmentation studies. This dataset has 2760 wound images in total, consists of wounds captured from different lighting, backgrounds, and angles. This variation makes the dataset useful for understanding general wound appearance and image quality differences. As this dataset does not provide Infection/ Inflammation, Moisture, and Edge conditions labels for the wounds, the images are all treated as unlabeled data in this study. This dataset is mainly used to support exploratory analysis and model development.

Besides, the private dataset consists of around 71 chronic wound images collected through collaboration with clinicians from clinical settings. These images are annotated by clinicians based on the TIME framework and include labels for Infection/ Inflammation, Moisture, and Edge conditions. As this manual annotation data collection needs clinical judgement and experience, the collected data is limited. This reflects

common challenges in clinical data collection. The labelled private dataset is therefore used as the main source for training and evaluating the IME-ClassNet model.

3.3 Data Preprocessing

All wound images, including publicly available datasets and private datasets, will undergo a series of preprocessing steps to ensure consistency and to readily prepare for analysis. This step helps to reduce variation caused by image capture conditions and to allow fair comparison during model training.

A brief manual quality check is performed on the public dataset to remove images with severe quality issues, such as extreme blur or incomplete wound visibility. Some of the wounds have multiple images from different angles. All the images were grouped by independent wound, which helps to prevent data leakage when both images are assigned to both the training and validation groups later.

3.3.1 Image Standardization

First, all images are resized to 224 x 224 pixels as required by the IME-ClassNet model. Bilinear interpolation is used to resize images because it provides smooth image scaling without having sharp artefacts. Pixel values are then normalized to the range [0, 1] by dividing by 255 to reduce the effect of lighting differences and to support efficient model learning. The color information is preserved, and images are processed in RGB format. This is because color cues are important for identifying Infection or Inflammation, Moisture, and Edge condition of a wound.

3.3.2 Region of Interest (ROI) Extraction

Next, the wound images are passed through the tissue segmentation model (T-SegNet) to generate the wound masks. The generated mask helps to identify the exact point of the wound from the wound images and this will aid in the ROI cropping process. The

segmentation mask identifies the wound area and is used to define the exact ROI for cropping. The wound region is enclosed within a bounding area derived from the segmentation output, and a small margin around the wound will be retained during cropping. This is to ensure the inclusion of periwound skin, which contains the important parts for assessing the wound edge. This step is to improve the reliability and consistency of IME classification and reduce noise from the surrounding unrelated background.

3.4 Exploratory Data Analysis (EDA)

EDA is performed on both public and private datasets to gain a basic understanding of the data and guide modelling decisions.

3.4.1 IME Class Distribution

For the private clinical dataset, EDA was used to explore the distribution of IME labels. As shown in **Figure 3.0**, the dataset shows significant class imbalance across all 3 features. In particular, the graph for the Moisture task skewed towards ‘Moisture’ class (N = 46), which dominates the minority classes (‘Too_Dry’ and ‘Too_Wet’). The class distribution patterns observed in **Figure 3.0** will affect the selection of the Class Weighting strategy and suitable evaluation metrics..

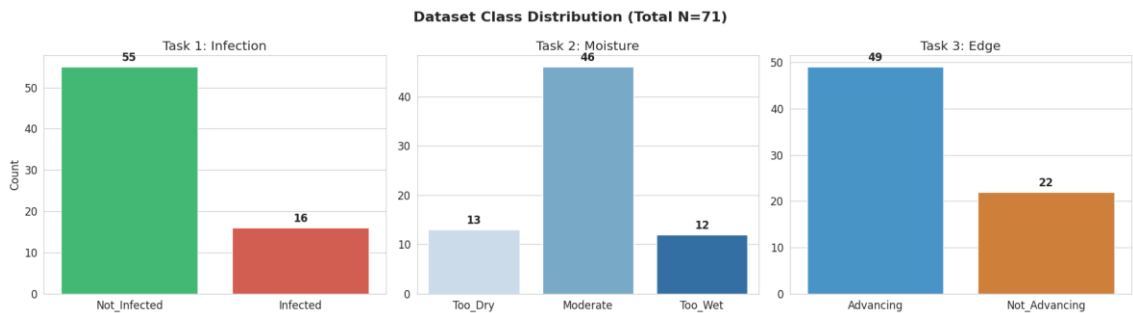


Figure 3.0: Class Distribution for Infection, Moisture, and Edge tasks in the labeled dataset (N = 71).

3.4.2 Image Quality and Variability

EDA is also used to assess variation in image quality and capture conditions, such as differences in lighting, background, distance, and viewing angle. This inspection helps confirm the need for preprocessing steps such as normalisation, ROI extraction, and data augmentation during model training. Images with poor quality will be excluded through the quality filtering process as described in Section 3.3.

3.5 Tissue Segmentation (T-SegNet)

Tissue segmentation is an important step in the proposed TIME-Net framework. It functions to identify the wound region accurately from the wound images. In this study, tissue segmentation is used to support the IME classification by locating the wound area and guiding the ROI extraction.

3.5.1 Role of T-SegNet

The tissue segmentation model used in this study is known as T-SegNet. It is developed by a collaborator and provided as a pre-trained model. The input will be all the RGB wound images, and the output will provide the segmented wound mask and a pixel-level predictions of different types of tissue including Granulation, Slough, and Necrotic Tissue. For the purpose of this study, the segmented wound mask will help to distinguish the wound area from the background.

3.5.2 Integration and Scope

The T-SegNet model is treated as a fixed upstream component that produces binary wound masks. The internal architecture, training process, and optimization of T-SegNet are not part of this research. The segmentation output is only used to support ROI extraction for IME classification.

3.6 IME-ClassNet Model Architecture

The main contribution of this study is the IME-ClassNet, which is a unified deep Convolutional Neural Network (CNN) designed to predict the classification of the 3 wound features: Infection, Moisture, and Edge condition. A transfer learning approach is adopted by utilizing a Multi-Task Learning (MTL) architecture based on a lightweight convolutional backbone. The goal is to achieve high inference efficiency within a smartphone-based wound assessment workflow with accurate diagnostics.

3.6.1 Multi-Task Learning Design

IME-ClassNet uses a shared feature extraction approach, where a single backbone network learns common visual features and branches into separate task-specific output heads. There are 2 benefits of this design. It helps improve computational efficiency, as all 3 IME classification tasks are predicted in a single forward pass, to reduces memory usage and inference time, which is important for smartphone-based workflows with cloud-hosted inference. Next, the model is able to capture shared information and learn these tasks simultaneously as the IME attributes are clinically related and often share the same signs. This can help to reduce overfitting risk, especially when there is limited data.

3.6.2 Convolutional Backbone Network

The shared feature extractor in IME-ClassNet is based on the EfficientNet-B0 architecture. EfficientNet-B0 will process the preprocessed wound image data through convolutional layers to extract visible features, which are related to color, texture, and shape. Global average pooling approach is used to condense the final feature map into a compact feature representation. This shared feature representation is then used as the input for the task-specific classification branches.

3.6.3 Task-Specific Classification Heads

IME-ClassNet includes three independent classification heads, each corresponding to one component of the IME framework. The inflammation and edge heads perform binary classification to predict whether the wound is infected or not infected, and whether the advancing or not advancing of the edge, respectively. At the same time, the moisture head performs multi-class classification to categorize the wound exudate level into too dry, moderate, and too wet.

While each classification head generates its own task-specific output, they all receive the same shared feature representation. Each wound image's final IME assessment is generated using the probability scores from these outputs.

3.7 Model Training

The IME-ClassNet model is trained using the preprocessed wound images. The model training pipeline is designed to make effective use of the limited labeled dataset while reducing the risk of overfitting and bias towards majority classes.

3.7.1 Supervised Transfer Learning and Fine-Tuning

Transfer learning approach is applied to the IME-ClassNet model to improve training efficiency under limited labelled data conditions. The EfficientNet-B0, which was pre-trained on the ImageNet dataset, is selected as to serve as the backbone network. This allows the model to learn basic feature detectors such as edges, textures, and color patterns, as the starting point for wound analysis.

The core backbone of EfficientNet-B0 is frozen during the first training phase, and only the task-specific classification heads are trained. This allows the model to adapt the IME classification tasks without changing its general visual representation. After that, the

top layers of the EfficientNet-B0 backbone are gradually “unfrozen” and fine-tuned using a lower learning rate to allow further adaptation to wound-specific features.

3.7.2 Data Augmentation

The limited training set is applied with data augmentation techniques such as random rotations, flipping, zooms, cropping, and lighting adjustments to increase the size for training, which also replicates the real-world variations in camera angles, to achieve model generalizability (Rishi Kesaraju, 2025).

3.7.3 Handling Class Imbalance

Class-weighted loss technique is used to reduce the bias of the model towards majority class. The loss weights are assigned according to the relative frequency of each IME category. With this, the model is encouraged to gain balanced representations of all components, including conditions that appear less frequent.

3.7.4 Optimization Configuration

Model optimization is performed using the Adam optimizer. Training is performed in small batches to manage memory usage efficiently and will be stopped when validation performance no longer improves.

3.8 Semi-Supervised Learning

Semi-supervised learning approach based on a pseudo-labelling approach is proposed to overcome the limitations of the small labelled dataset while utilizing the large unlabeled public dataset. This method makes use of the extra wound images from unlabeled dataset without requiring manual annotation (Liu et al., 2023).

3.8.1 Baseline Model Training

A baseline IME-ClassNet model is trained using the labelled private clinical dataset. This training follows the procedure in Section 3.7. Although this baseline model is trained

on a small labeled dataset, it provides an initial IME classifier that captures the fundamental characteristics of IME from wound images. This will serve as the starting point for generating predictions on unlabeled data in the next phase.

3.8.2 Pseudo-Label Generation and Filtering

The large unlabeled public dataset will generate the IME classification prediction using the baseline model. A confidence-based filtering process is applied to filter out those with low confidence results. This is to avoid introducing incorrect labels into the training set later. Only predictions with a high confidence rate will be treated as pseudo-labels. This filtering process allows reliable unlabeled samples to contribute additional training information and subsequently increases the sample size in future training. The selected pseudo-labelled images will then be combined with the original labeled dataset to form an expanded training set.

3.8.3 Model Retraining

A new instance of the IME-ClassNet model is trained using the expanded dataset that includes both original labeled images and the selected pseudo-labeled images. This newly combined dataset will be used to train a new model, which starts with pre-trained ImageNet weights and training using the same procedure as described in Section 3.7.

The new model will be exposed to more wound appearances, lighting conditions, and image capturing angles, which subsequently improve the model generalizability.

3.9 Model Evaluation

The IME-ClassNet model performance is evaluated using the metrics below:

3.9.1 Stratified k-fold Cross-validation

As this study has limited labelled wound images, a traditional train-test split is not appropriate to be used in this study. On the other hand, this study uses a stratified k-fold

cross-validation method to test model performance. This method allows all the images to be used for both training and validation across different folds.

In this study, the newly labeled dataset is divided into 5 folds while maintaining the distribution of IME labels across folds. For each round, one fold is used for validation, and the other 4 folds will be used for training. This process is repeated until every fold has been used once for validation, and the average across 5 folds are reported as the final model performance.

3.9.2 Evaluation Metrics

IME-ClassNet model performance is evaluated using standard classification metrics such as F1-score, Sensitivity (Recall), and Precision. The standard accuracy is not use as the primary metric because of the class imbalance of the 3 features.

The F1-score provides a balanced measure of classification performance by using the mean of Precision and Sensitivity (Recall), making it suitable to be used for a class imbalance dataset.

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Sensitivity measures the ability to identify positive cases, such as infection or not advancing edges.

$$Sensitivity = \frac{TP}{TP + FN}$$

Precision measures the reliability of positive predictions. This helps to check for the rate of False predictions.

$$Precision = \frac{TP}{TP + FP}$$

3.9.3 Reporting

For each cross-validation fold, evaluation metrics are calculated on the validation data. Final model performance is reported as the mean and standard deviation across all folds to reflect both accuracy and stability. Confusion matrices are used to analyse common misclassification patterns across the IME features.

3.10 Model Deployment and System Evaluation

The proposed TIME-Net framework is then integrated into a smartphone-based prototype workflow for near real-time wound assessment.

3.10.1 Model Deployment Framework

The trained IME-ClassNet together with the T-SegNet model will be combined as TIME-Net and integrated into a smartphone-based prototype. The prototype is designed to accept wound images captured using smartphone devices as input, then perform preprocessing and model inference, and display the tissue segmentation and IME classification results to the user. The smartphone-based prototype is intended as a proof-of-concept system to demonstrate feasibility rather than a fully commercial product.

3.10.2 End-to-End Performance Benchmarking

In the prototype, inference is performed through a managed cloud inference service (remote inference), while the smartphone device functions as the image capture and result display interface. This design simplifies deployment and careful handling of wound images to address privacy and governance considerations. Performance is evaluated using end-to-end response time, measured from the moment an image is submitted until the full TIME outputs (mask + IME predictions) are returned and displayed, including network transmission and server-side inference time. Where applicable, the total response time can be reported as the sum of (i) T-SegNet inference time, (ii) ROI extraction and preprocessing time, and (iii) IME-ClassNet inference time.

3.10.3 Performance Target and Evaluation Criteria

End-to-end response time is measured and analyzed to assess the deployment feasibility of the proposed cloud-based TIME-Net framework, which is considered acceptable for practical use in clinical or home-based wound monitoring scenarios. Evaluation is performed by repeating inference runs across multiple wound images and reporting the mean response time.

3.11 Summary

This chapter has presented the methodology used for developing and evaluating the proposed TIME-Net framework for image-based chronic wound assessment. This study utilizes 2 datasets: a publicly available unlabeled large dataset and a private clinical labeled small dataset. All the wound images will undergo data preparation, including image standardization, region-of-interest extraction guided by tissue segmentation, and quality filtering. A pre-trained tissue segmentation (T-SegNet) developed by collaborator will be used to support the IME classification prediction. The IME-ClassNet architecture was designed as a multi-task learning with lightweight pre-trained backbone, which suitable for mobile deployment.

Transfer learning, data augmentation, handling class imbalance, and optimization considerations were included in the model training section. A semi-supervised learning approach based on pseudo-labelling was introduced to overcome the small dataset limitation. The model will be assessed using clinically relevant metrics like F1-score, Sensitivity, and Precision as well as a stratified cross-validation technique. The end-to-end response time using cloud-hosted inference will be used to assess the practicality of implementing the proposed framework on a smartphone device.

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